20261111 Quiz

for (MATH200) Differential Equations

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Problem 6.1.6. Use the linearity of the Laplace transform to find the Laplace transform of

$$f(t) = e^{at}\cosh(bt),$$

where a and b are real constants.

Solution. Using $\cosh(bt) = (e^{bt} + e^{-bt})/2$, we have

$$f(t) = \frac{1}{2}e^{(a+b)t} + \frac{1}{2}e^{(a-b)t}.$$

By linearity,

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \frac{1}{2}\left(\frac{1}{s-a-b} + \frac{1}{s-a+b}\right) \\ &= \boxed{\frac{s-a}{(s-a)^2 - b^2}}, \qquad s > a + |b|.\end{aligned}$$

□

Problem 6.1.7. Use the linearity of the Laplace transform to find the Laplace transform of

$$f(t) = e^{at}\sinh(bt),$$

where a and b are real constants.

Solution. Since $\sinh(bt) = (e^{bt} - e^{-bt})/2$, linearity gives

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \frac{1}{2}\left(\frac{1}{s-a-b} - \frac{1}{s-a+b}\right) \\ &= \boxed{\frac{b}{(s-a)^2 - b^2}}, \qquad s > a + |b|.\end{aligned}$$

□

Problem 6.1.10. Use the linearity of the Laplace transform to find the Laplace transform of

$$f(t) = e^{at}\cos(bt),$$

where a and b are real constants.

Solution. Euler's formula yields

$$e^{at}\cos(bt) = \frac{1}{2}\left(e^{(a+ib)t} + e^{(a-ib)t}\right).$$

Thus, using the exponential transform and combining the conjugate terms,

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \frac{1}{2}\left(\frac{1}{s-a-ib} + \frac{1}{s-a+ib}\right)\\ &= \boxed{\frac{s-a}{(s-a)^2+b^2}}, \qquad s > a.\end{aligned}$$

□

Problem 6.1.11. Use the linearity of the Laplace transform to find the Laplace transform of

$$f(t) = e^{at}\sin(bt),$$

where a and b are real constants.

Solution. Using $\sin(bt) = (e^{ibt} - e^{-ibt})/(2i)$, we obtain

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \frac{1}{2i}\left(\frac{1}{s-a-ib} - \frac{1}{s-a+ib}\right)\\ &= \boxed{\frac{b}{(s-a)^2+b^2}}, \qquad s > a.\end{aligned}$$

□

Problem 6.1.18. Find the Laplace transform of the function

$$f(t) = \begin{cases} t, & 0 \le t < 3,\\ 1, & t \ge 3.\end{cases}$$

Solution. The definition of the Laplace transform gives

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \int_0^3 te^{-st}\,\mathrm{d}t + \int_3^\infty e^{-st}\,\mathrm{d}t\\ &= \frac{1-(1+3s)e^{-3s}}{s^2} + \frac{e^{-3s}}{s}.\end{aligned}$$

Consequently,

$$\boxed{\mathcal{L}\{f(t)\}(s) = \frac{1-(1+2s)e^{-3s}}{s^2}}, \qquad s > 0.$$

□

Problem 6.2.2. Find the inverse Laplace transform of

$$F(s) = \frac{5}{(s-1)^3}.$$

Solution. Since $\mathcal{L}\{t^2e^t\}(s) = 2/(s-1)^3$, we get

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = \frac{5}{2}t^2e^t}.$$

□

Problem 6.2.6. Find the inverse Laplace transform of

$$F(s) = \frac{8s^2 - 6s + 12}{s(s^2+4)}.$$

Solution. The partial fraction decomposition is

$$\frac{8s^2 - 6s + 12}{s(s^2+4)} = \frac{3}{s} + \frac{5s-6}{s^2+4}.$$

Hence

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = 3 + 5\cos(2t) - 3\sin(2t)}.$$

□

Problem 6.2.7. Find the inverse Laplace transform of

$$F(s) = \frac{3-2s}{s^2+4s+5}.$$

Solution. Completing the square and rearranging the numerator,

$$F(s) = \frac{-2(s+2)+7}{(s+2)^2+1}.$$

Thus

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = e^{-2t}(-2\cos t + 7\sin t)}.$$

□

Problem 6.2.14. Use the Laplace transform to solve the initial value problem

$$y^{(4)} - 4y''' + 6y'' - 4y' + y = 0,$$
$$y(0) = 0, \qquad y'(0) = 1, \qquad y''(0) = 0, \qquad y'''(0) = 1.$$

Solution. Let $Y(s) = \mathcal{L}\{y(t)\}(s)$. Taking transforms and using all four initial conditions gives

$$(s^4 - 4s^3 + 6s^2 - 4s + 1)Y(s) = s^2 - 4s + 7.$$

Since the polynomial on the left is $(s-1)^4$,

$$\begin{aligned} Y(s) &= \frac{s^2 - 4s + 7}{(s-1)^4} \\ &= \frac{1}{(s-1)^2} - \frac{2}{(s-1)^3} + \frac{4}{(s-1)^4}. \end{aligned}$$

Taking inverse transforms,

$$\boxed{y(t) = e^t\left(t - t^2 + \frac{2}{3}t^3\right)}.$$

This expression satisfies the differential equation and the four prescribed initial conditions.

□

Problem 6.2.15. Use the Laplace transform to solve the initial value problem

$$y'' - 2y' + 2y = \cos t, \qquad y(0) = 1, \qquad y'(0) = 1.$$

Solution. Taking Laplace transforms yields

$$(s^2 - 2s + 2)Y(s) = s - 1 + \frac{s}{s^2 + 1}.$$

Partial fractions give

$$Y(s) = \frac{4(s-1) + 3}{5((s-1)^2 + 1)} + \frac{s - 2}{5(s^2 + 1)}.$$

Therefore

$$\boxed{y(t) = \frac{1}{5}\left(e^t(4\cos t + 3\sin t) + \cos t - 2\sin t\right)}.$$

The solution satisfies both initial conditions and the given differential equation.

□

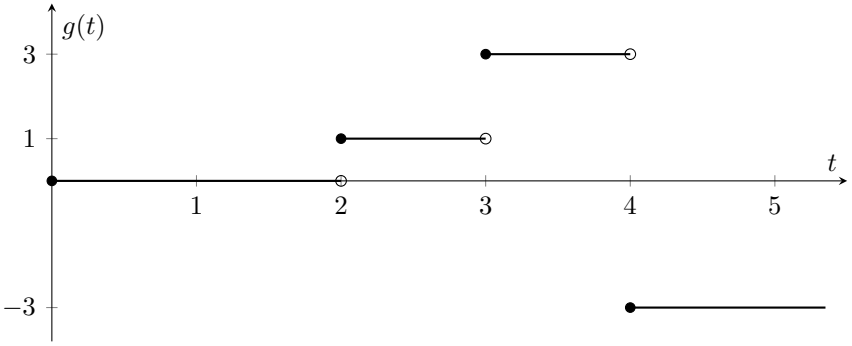
Problem 6.3.1. Sketch the graph of the function

$$g(t) = u_2(t) + 2u_3(t) - 6u_4(t), \qquad t \geq 0.$$

Solution. Here $u_c(t) = H(t-c)$ denotes the unit step function, with $H(0) = 1$ as in the textbook. The value of g changes at $t = 2, 3, 4$, and

$$g(t) = \begin{cases} 0, & 0 \leq t < 2, \\ 1, & 2 \leq t < 3, \\ 3, & 3 \leq t < 4, \\ -3, & t \geq 4. \end{cases}$$

The filled and open circles indicate included and excluded endpoint values, respectively.



□

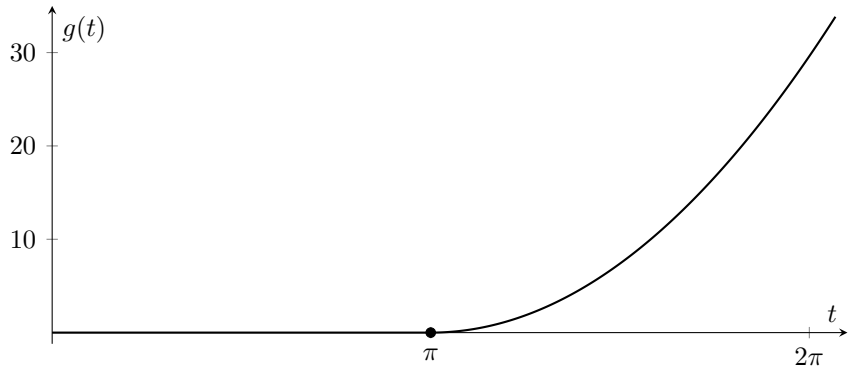
Problem 6.3.2. Sketch the graph of the function

$$g(t) = f(t - \pi)u_\pi(t), \qquad f(t) = 3t^2, \qquad t \geq 0.$$

Solution. Because $u_\pi(t) = H(t-\pi)$, the function is

$$g(t) = \begin{cases} 0, & 0 \le t < \pi, \\ 3(t-\pi)^2, & t \ge \pi. \end{cases}$$

Thus the graph coincides with the t-axis until $t = \pi$ and then follows an upward-opening parabola with vertex $(\pi, 0)$.



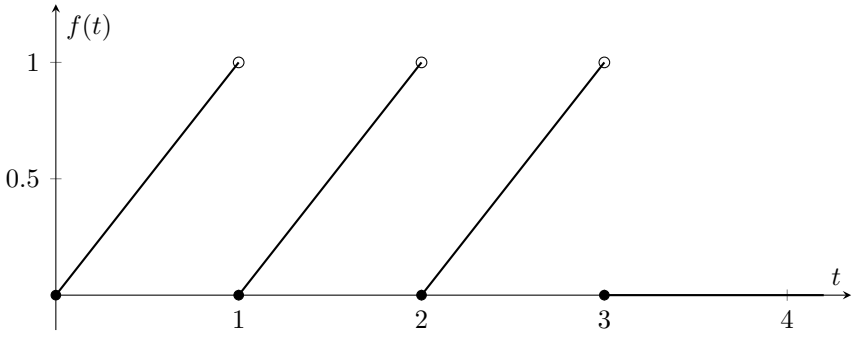
□

Problem 6.3.7. Consider the function

$$f(t) = \begin{cases} t, & 0 \le t < 1, \\ t-1, & 1 \le t < 2, \\ t-2, & 2 \le t < 3, \\ 0, & t \ge 3. \end{cases}$$

(a) Sketch the graph of f.
(b) Express $f(t)$ in terms of the unit step function $u_c(t)$.

Solution. (a) On each of the intervals $[0,1)$, $[1,2)$, and $[2,3)$, the graph is a line segment of slope 1. The function jumps from the left-hand limit 1 to 0 at $t = 1, 2, 3$.



(b) Starting from t, subtract 1 at both $t = 1$ and $t = 2$, and then subtract $t - 2$ for $t \ge 3$. Thus

$$\boxed{f(t) = t - u_1(t) - u_2(t) - (t-2)u_3(t)}.$$

Equivalently, since $u_c(t) = H(t-c)$,

$$f(t) = t - H(t-1) - H(t-2) - (t-2)H(t-3).$$

□

Problem 6.3.9. Find the Laplace transform of

$$f(t) = \begin{cases} 0, & t < 2, \\ (t-2)^3, & t \geq 2. \end{cases}$$

Solution. Since $f(t) = H(t-2)(t-2)^3$, the second shifting theorem gives

$$\mathcal{L}\{f(t)\}(s) = e^{-2s}\mathcal{L}\{t^3\}(s) = e^{-2s}\frac{3!}{s^4}.$$

Therefore

$$\boxed{\mathcal{L}\{f(t)\}(s) = \frac{6e^{-2s}}{s^4}}, \qquad s > 0.$$

□

Problem 6.3.15. Find the inverse Laplace transform of

$$F(s) = \frac{(s-2)e^{-2s}}{s^2 - 4s + 3}.$$

Solution. Complete the square in the denominator:

$$s^2 - 4s + 3 = (s-2)^2 - 1.$$

Since

$$\mathcal{L}\{e^{2t}\cosh t\}(s) = \frac{s-2}{(s-2)^2 - 1},$$

the factor e^{-2s} produces a shift by 2 in time. Therefore

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = H(t-2)e^{2(t-2)}\cosh(t-2)}.$$

□

Problem 6.3.21. Find the Laplace transform of the function

$$f(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 0, & 1 \leq t < 2, \\ 1, & 2 \leq t < 3, \\ 0, & t \geq 3. \end{cases}$$

Solution. The function is nonzero only on $[0, 1)$ and $[2, 3)$. Hence

$$\begin{aligned} \mathcal{L}\{f(t)\}(s) &= \int_0^1 e^{-st}\,\mathrm{d}t + \int_2^3 e^{-st}\,\mathrm{d}t \\ &= \frac{1 - e^{-s}}{s} + \frac{e^{-2s} - e^{-3s}}{s}. \end{aligned}$$

Thus

$$\boxed{\mathcal{L}\{f(t)\}(s) = \frac{1 - e^{-s} + e^{-2s} - e^{-3s}}{s}}, \qquad s > 0.$$

□

Problem 6.4.2. Consider the initial value problem

$$y'' + y = g(t), \qquad y(0) = 0, \qquad y'(0) = 2,$$

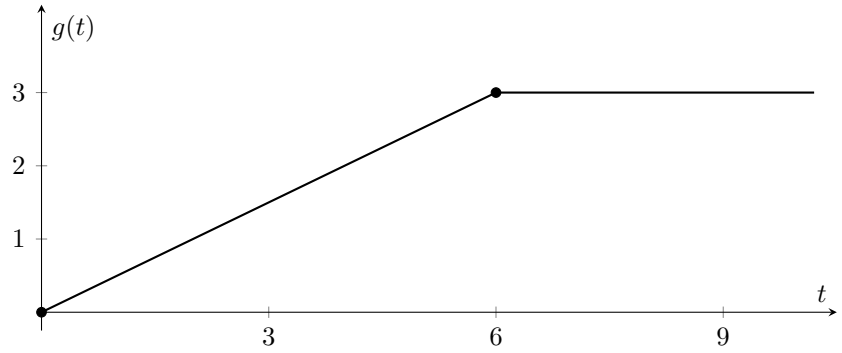
where

$$g(t) = \begin{cases} t/2, & 0 \le t < 6, \\ 3, & t \ge 6. \end{cases}$$

(a) Sketch the graph of the forcing function on an appropriate interval.

(b) Find the solution of the initial value problem.

Solution. (a) The forcing function increases linearly from $g(0) = 0$ to $g(6) = 3$, and is constant thereafter. It is continuous at $t = 6$.



(b) Since $g(t) = t/2$ for $t < 6$ and then remains at 3, it can be written as

$$g(t) = \frac{t}{2} - \frac{1}{2}H(t-6)(t-6).$$

Taking Laplace transforms and using the initial conditions gives

$$(s^2+1)Y(s) - 2 = \frac{1 - e^{-6s}}{2s^2},$$

so

$$Y(s) = \frac{2}{s^2+1} + \frac{1 - e^{-6s}}{2s^2(s^2+1)}.$$

The partial fraction decomposition

$$\frac{1}{s^2(s^2+1)} = \frac{1}{s^2} - \frac{1}{s^2+1}$$

shows that its inverse transform is $h(t) = t - \sin t$. Applying the second shifting theorem to the term containing e^{-6s} yields

$$\boxed{y(t) = \frac{3}{2}\sin t + \frac{t}{2} - \frac{1}{2}H(t-6)(t-6-\sin(t-6))}.$$

For $0 \le t < 6$, this gives $y'' + y = t/2$. For $t > 6$, the shifted term changes the forcing to 3. The solution satisfies $y(0) = 0$ and $y'(0) = 2$, and both y and y' are continuous at $t = 6$.

□

Problem 6.5.2. Solve the initial value problem

$$y'' - y = -5\delta(t-3), \qquad y(0) = 1, \qquad y'(0) = 0.$$

Solution. Taking Laplace transforms gives

$$(s^2-1)Y(s) - s = -5e^{-3s},$$

so

$$Y(s) = \frac{s}{s^2-1} - \frac{5e^{-3s}}{s^2-1}.$$

Using $\mathcal{L}^{-1}\{s/(s^2-1)\}(t) = \cosh t$ and $\mathcal{L}^{-1}\{1/(s^2-1)\}(t) = \sinh t$, we obtain

$$\boxed{y(t) = \cosh t - 5H(t-3)\sinh(t-3)}.$$

The impulse at $t = 3$ leaves y continuous and produces the jump $y'(3^+) - y'(3^-) = -5$, as required by the equation.

□

Problem 6.6.4. Find the Laplace transform of

$$f(t) = \int_0^t (t-\tau)^2 \cos(3\tau)\ \mathrm{d}\tau.$$

Solution. With $p(t) = t^2$ and $q(t) = \cos(3t)$, the given function is $f = p * q$. By the convolution theorem,

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \mathcal{L}\{p(t)\}(s)\mathcal{L}\{q(t)\}(s)\\ &= \frac{2}{s^3}\cdot\frac{s}{s^2+9}.\end{aligned}$$

Therefore

$$\boxed{\mathcal{L}\{f(t)\}(s) = \frac{2}{s^2(s^2+9)}}, \qquad s > 0.$$

□

Problem 6.6.5. Find the Laplace transform of

$$f(t) = \int_0^t e^{-(t-\tau)}\sin(2\tau)\ \mathrm{d}\tau.$$

Solution. Let $p(t) = e^{-t}$ and $q(t) = \sin(2t)$. Then $f = p * q$, so

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \mathcal{L}\{p(t)\}(s)\mathcal{L}\{q(t)\}(s)\\ &= \frac{1}{s+1}\cdot\frac{2}{s^2+4}.\end{aligned}$$

Hence

$$\boxed{\mathcal{L}\{f(t)\}(s) = \frac{2}{(s+1)(s^2+4)}}, \qquad s > 0.$$

□

Problem 6.6.6. Find the Laplace transform of

$$f(t) = \int_0^t \sin(t-\tau)\cos(2\tau)\ \mathrm{d}\tau.$$

Solution. Here $p(t) = \sin t$ and $q(t) = \cos(2t)$, so $f = p * q$. Thus

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= \mathcal{L}\{p(t)\}(s)\mathcal{L}\{q(t)\}(s) \\ &= \frac{1}{s^2+1}\cdot\frac{s}{s^2+4}.\end{aligned}$$

Therefore

$$\boxed{\mathcal{L}\{f(t)\}(s) = \frac{s}{(s^2+1)(s^2+4)}}, \qquad s > 0.$$

□

Problem 6.6.8. Find the inverse Laplace transform of

$$F(s) = \frac{1}{s^4(s^2+4)}$$

by using the convolution theorem.

Solution. Factor the transform as

$$F(s) = \frac{1}{s^4}\cdot\frac{1}{s^2+4}.$$

The inverse transforms of the factors are

$$\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}(t) = \frac{t^3}{6}, \qquad \mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\}(t) = \frac{1}{2}\sin(2t).$$

By the convolution theorem,

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{12}\int_0^t (t-\tau)^3\sin(2\tau)\ \mathrm{d}\tau}.$$

□

Problem 6.6.9. Find the inverse Laplace transform of

$$F(s) = \frac{s}{(s+1)(s^2+9)}$$

by using the convolution theorem.

Solution. Write

$$F(s) = \frac{1}{s+1}\cdot\frac{s}{s^2+9}.$$

The factors are the Laplace transforms of e^{-t} and $\cos(3t)$, respectively. Hence

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = \int_0^t e^{-(t-\tau)}\cos(3\tau)\ \mathrm{d}\tau}.$$

□

Problem 6.6.10. Find the inverse Laplace transform of

$$F(s) = \frac{1}{(s+1)^3(s^2+4)}$$

by using the convolution theorem.

Solution. We factor

$$F(s) = \frac{1}{(s+1)^3} \cdot \frac{1}{s^2+4},$$

and use the inverse transforms

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^3}\right\}(t) = \frac{1}{2}t^2e^{-t}, \qquad \mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\}(t) = \frac{1}{2}\sin(2t).$$

Thus

$$\boxed{\mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{4}\int_0^t (t-\tau)^2 e^{-(t-\tau)}\sin(2\tau)\ \mathrm{d}\tau}.$$

□

Problem 6.6.14. Express the solution of the initial value problem

$$4y'' + 4y' + 17y = g(t), \qquad y(0) = 0, \qquad y'(0) = 1$$

in terms of a convolution integral.

Solution. Let $Y(s) = \mathcal{L}\{y(t)\}(s)$ and $G(s) = \mathcal{L}\{g(t)\}(s)$. The initial conditions yield

$$(4s^2 + 4s + 17)Y(s) = G(s) + 4.$$

Since

$$4s^2 + 4s + 17 = 4\left((s+\tfrac{1}{2})^2 + 4\right),$$

we obtain

$$Y(s) = \frac{1}{(s+\frac{1}{2})^2+4} + \frac{G(s)}{4((s+\frac{1}{2})^2+4)}.$$

The first term has inverse transform $\frac{1}{2}e^{-t/2}\sin(2t)$. The impulse-response kernel corresponding to the second term is

$$h(t) = \mathcal{L}^{-1}\left\{\frac{1}{4((s+\frac{1}{2})^2+4)}\right\}(t) = \frac{1}{8}e^{-t/2}\sin(2t).$$

The convolution theorem now gives

$$\boxed{y(t) = \frac{1}{2}e^{-t/2}\sin(2t) + \frac{1}{8}\int_0^t e^{-(t-\tau)/2}\sin(2(t-\tau))g(\tau)\ \mathrm{d}\tau}.$$

□